

Response to the Reviewers

Open-Source Cross-Domain Datasets Enabled by LLM-Driven Digital Twins and Network Telemetry

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We thank the reviewers for their careful and constructive assessment. We reorganised the article around practical and generated datasets, clarified the contribution of the cross-domain architecture, expanded the experimental methodology, and added reproducibility and correctness checks. Detailed execution records and file-level metadata are provided in the public revision artefacts.

The principal revisions are:

1. We clarify that cross-domain describes the implemented system architecture and unified data-interface workflow, while platform scheduling required the practical datasets to be collected in different time periods.
2. We apply CC BY-NC 4.0 to the released datasets and associated metadata/documentation, use “openly available” consistently, and state the licence in Table 2 and Croissant metadata.
3. We define the multi-agent workflow operationally through schema-defined roles, shared orchestrator state, simulator tools, and retained intermediate artefacts, and evaluate it through an instrumented run and a deterministic baseline.
4. We add leakage-free temporal semantic evaluation, multiple privacy attackers and baselines, a deterministic GNPY enumerator, versioned GNPY replay, and bounded ns-3 validation.
5. We remove the template DOI and compilation date, rebuild all citations and references, enlarge Fig. 6, regenerate Fig. 7, and standardise terminology and British English.

Text shown in green describes our response; reviewer comments are shown in blue. Repository paths are retained where they are the stable names of released artefacts.

Reviewer 1

The paper addresses a relevant problem and presents a good collection of datasets and tools

A few comments to improve the quality

- The “cross-domain” claim is not really clear, the optical, sensing, and wireless datasets seem to originate from separate measurement campaigns. Please clarify

Response. We thank the reviewer for this comment. Cross-domain refers to our system architecture and data-processing workflow. The framework connects optical transport, fibre sensing, and wireless access through common repository, semantic-processing, and training interfaces, and these functions can be invoked across the supported domains. Because of experimental-platform usage and scheduling constraints, the practical datasets had to be collected in different time periods. This is an acquisition condition rather than a limitation of the cross-domain architecture. In subsequent training, our unified data interface loads each source, invokes the corresponding domain-specific preprocessing, and supplies the optical and wireless features to the same learning workflow.

Change in the manuscript. The Abstract, Introduction, Section III, Section IV-B, the practical dataset description, and the Conclusions now distinguish the cross-domain system architecture from the different acquisition periods and describe the unified data-interface calls used during training.

- Provide details about the LLM (model used, number of attempts, tool interface, failure rate, cost, latency, or degree of human intervention). Would it be feasible a conventional scripting pipeline as baseline?

Response. We now position the LLM workflow as a feasibility demonstration. A separately labelled instrumented run used `gpt-4o-mini-2024-07-18`, temperature 0.2, a 90-s timeout, at most three attempts per call, and strict mode without silent rule fallback. Planner, Scenario Expansion, and Reflection were each called twice. All six API calls succeeded on their first attempt, consuming 106,041 input and 2,267 output tokens (108,308 total), with 26.22 s summed API latency and an estimated text-token cost of US\$0.01727 using the dated price snapshot stored with the run.

Manual recovery was required during simulator execution. An earlier invocation made two successful API calls (4,057 tokens; estimated US\$0.0009339) before it was stopped after 344 GNPY import failures and one all-off result. Recovery included installing the local package, correcting inactive-channel representation and result extraction, repairing a single-channel case, resuming from manifest-hashed files, correcting the recovery log callback, and rerunning nine failed scenarios. A later deterministic replay corrected a power-sweep override and reused three saved LLM responses, so it incurred no new API calls. These events are reported here and in the public trace; the article states only that manual recovery was required.

The independent script enumerates the expected Cartesian product of four destinations, two modulation/slot-width settings, 256 occupancy patterns, and three powers. It generates 6,144 unique configuration keys without an LLM or simulator execution. The LLM-orchestrated history contains 8,192 execution records but the same 6,144 unique keys, hence a 25% repetition rate and set precision/recall of 1.0. We do not compare enumeration-only time with end-to-end LLM plus GNPY runtime and make no superiority claim.

Change in the manuscript. Section V-B and Tables 3–4 report the model/settings, observed calls, usage, cost, API latency, manual-recovery boundary, and deterministic baseline. Detailed chronology, hashes, requests, responses, and failure logs remain in the revision repository.

Evidence and boundary. The trace contains six rendered requests, six complete responses, per-call records, prompts, model identifiers, timings, token usage, source/input hashes, and recovery history. This new run does not reconstruct the unavailable historical API trace and is not a repeated-trial reliability estimate.

- Would be nice to have a summary table for each dataset reporting the number of samples, features, labels, file format, storage size, licence... You may also consider to align with <https://mlcommons.org/working-groups/data/croissant/>

Response. We added a dataset summary table and a Croissant 1.0-compatible JSON-LD record. The table distinguishes the two 339-km Voyager campaigns, the 986-km and urban-sensing sources, wireless telemetry, the constructed benchmark, and both synthetic datasets. It reports record/sampling information, principal fields or labels, formats/sizes, and access/reuse terms.

We cross-checked the practical-data README, Croissant record, Table 2, and manuscript text. The 2023–24 Voyager product contains 1,373 processed 15-minute records. The distinct 2025–26 campaign contains 20,960 raw eight-channel records with real timestamps, mostly 70–72-s spacing and acquisition gaps, and was not resampled to 15-minute intervals.

We also resolved the licence terminology directly. The authors intend the datasets to be reusable for non-commercial research, so we added a repository-level data-licence notice applying Creative Commons Attribution–NonCommercial 4.0 International (CC BY-NC 4.0) to the released datasets and associated metadata/documentation. This licence permits sharing and adaptation with attribution for non-commercial purposes. Software and identified third-party material retain their own terms. Because the non-commercial condition is narrower than the formal open-source definition, the manuscript uses “openly available” rather than the potentially ambiguous “open-source” label.

Change in the manuscript. Table 2 and the practical dataset subsection now identify CC BY-NC 4.0 explicitly. The Abstract, Introduction, and Conclusions use “openly available”, and the Croissant `license` field contains the canonical licence URL. The original manuscript title is retained.

Evidence and boundary. The repository contains a root `LICENSE` notice; the practical-data release package links to the same terms. The Croissant record includes the licence, access locations, sizes, checksums, and field units where defined. Unknown external collection totals and undocumented raw wireless units are marked unavailable rather than inferred.

Reviewer 2

This paper presents an open-source framework for constructing a dataset that integrates the optical- and wireless-communication domains. The dataset comprises both real-world and synthetic data. Overall, the paper presents substantial work and covers a broad range of topics. However, the paper does not always distinguish the primary contributions from supporting material, some descriptions are unclear, and the boundary between this study and the authors’ previous work is insufficiently defined. The authors are requested to address the following comments and revise the manuscript accordingly.

1. The hierarchy of the contributions presented in the Introduction is unclear

The authors state that “this paper presents an open-source cross-domain dataset framework enabled by network telemetry, semantic data abstraction, and LLM-driven multi-agent DTs.” However, these three elements are not fully parallel. The first two relate to practical datasets, whereas LLM-driven multi-agent DTs relate to generated datasets. The contribution could instead be stated as follows: “This paper presents an open-source cross-domain dataset framework that supports both practical and generated datasets: the former through network telemetry and semantic data abstraction, and the latter through LLM-driven multi-agent DTs.”

Response. We adopted this hierarchy. Practical datasets are organised through telemetry and semantic abstraction; generated datasets are demonstrated through the LLM-driven DT workflow. The four contributions now follow this structure and distinguish inherited telemetry infrastructure from the new dataset organisation, semantic evaluation, DT demonstration, and reproducibility audit.

Change in the manuscript. The Abstract, final Introduction paragraphs, contribution list, and Conclusions were rewritten around practical versus generated data.

2. The Related Work section is not clearly organized

Subsections A, B, and C contain repeated content and arguments, conflicting statements, an unclear progression of ideas, and insufficient transitions. Some specific issues are described below:

1. The writing style and focus are inconsistent across the three subsections. Subsection A objectively summarizes existing studies without directly explaining the specific improvements introduced in this paper. Subsection B describes how previous work motivates and supports this study, but mentions the proposed improvement only in its final sentence. Subsection C objectively evaluates existing studies and briefly introduces the work related to this paper. The authors are advised to adopt a consistent structure: objectively assess the existing literature, identify its limitations, explain how it motivates or supports the present study where applicable, and then briefly introduce the proposed approach.

2. The opening statements of Subsections A, B, and C are not fully consistent. Subsection A argues that existing datasets are inadequate and proposes supplementing real-world data through telemetry. Subsection B discusses semantic extraction and privacy protection for real-world data. However, Subsection C begins by stating: “However, real network measurements are difficult to collect at scale because of limited testbed access, operational cost, privacy constraints, safety requirements, and the rarity of abnormal or extreme operating conditions. This challenge is more pronounced in cross-domain optical-wireless networks, where heterogeneous infrastructures, telemetry interfaces, and data formats complicate data collection and alignment.” This appears potentially self-contradictory. If the proposed work uses telemetry and, to some extent, addresses heterogeneous alignment and semantic extraction for real-world data, the quoted statement should be reconciled with those contributions. If the present work does not address these issues, the extensive discussion of these limitations in Subsections A and B appears less relevant. In addition, the transition from Subsections A and B to Subsection C needs to be strengthened.

3. The discussion of limitations in existing work is repetitive. The lack of uniformity between optical and wireless infrastructures and data characteristics, as well as the resulting alignment challenges, are discussed repeatedly in Subsections A and C. It would be preferable to complete this discussion in Subsection A. Subsection C could then focus on the value and importance of generated data, the limitations of related studies, and the solution proposed in this paper, without reopening the same cross-domain issues from the perspective of generated data. In addition, all three subsections mention that existing studies primarily address a single domain rather than cross-domain settings, while both Subsections A and B discuss data-privacy concerns. Each subsection should have a clearly defined primary focus and supporting scope so that the Related Work section develops progressively and distinguishes the main points from secondary ones.

4. The Introduction already presents the background, significance, and research content of this study. The authors should avoid turning the Related Work section into a more detailed, topic-by-topic version of the Introduction. Instead, this section should focus on the limitations of existing studies relative to the present work, the ways in which prior studies motivate or support this work, and the advances offered by the proposed approach.

Response. We reorganised the section into three functions: cross-domain data availability and telemetry; semantic abstraction and empirical privacy evaluation; and DT/LLM-generated data. Each subsection now describes representative work, identifies the remaining limitation, and closes with the corresponding advance in this study. Section II-C begins with the distinct value of controllable synthetic data rather than reopening the telemetry-alignment discussion.

We consolidated data availability and telemetry in Section II-A, information selection and privacy in Section II-B, and controllable scenario generation in Section II-C. The three subsections now have separate primary questions. We also shortened the Introduction to the problem, the two-part framework, and the contribution hierarchy.

Change in the manuscript. Section II was rewritten; repeated cross-domain/privacy arguments were removed; and the Introduction was shortened so it no longer duplicates the literature review.

5. The first sentence of the fourth paragraph in Section III states: “The orchestrator coordinates the interactions among multiple agents, DTs, AI engines, data repositories, and network controllers.” However, Fig. 1 appears to show only data-flow interactions between the orchestrator and the Agentic AI layer. How does the orchestrator coordinate or control the other components?

Response. We rewrote Section III to describe the implemented relationships directly. The orchestrator owns the shared task state and schedules planning, scenario expansion, DT execution, result analysis, optional reflection, and reporting; it exchanges structured requests and results with the role modules and invokes GNPY or ns-3 through simulator adapters. The AI engine and SDN controller are separate services inherited from our implemented optical telemetry platform [*JOCN*, 17(2), 139–150, 2025]. They consume repository data and prediction outputs, but the DT-workflow orchestrator does not dispatch those services.

Change in the manuscript. Section III and the Fig. 1 caption now identify the orchestrator’s control path, the simulator interface, and the connection to the previously implemented telemetry, AI-engine, and controller services.

6. Please add references to support the following statement in Section II-C: “Existing DT and LLM-agent workflows generally assume that simulation objectives, scenario spaces, parameter ranges, and data-generation requirements are manually specified. Limited work has considered dataset generation itself as an autonomous closed-loop workflow, especially across optical and wireless domains.”

Response. We replaced the unsupported generalisation with a comparison tied to cited DT assistants and AI-agent studies. Those works automate modelling or network operation but use supplied objectives or parameter ranges. We describe dataset generation as the workflow-design problem examined here and limit our own result to feasibility.

Change in the manuscript. Section II-C now cites the relevant GPT/DT assistant, LLM-DT management, local-agent, and DT-aided control studies immediately beside the bounded comparison.

7. The abstract states: “The proposed framework integrates distributed and centralised data repositories, Kafka-based telemetry pipelines, DTs, AI engines, and network controllers to support scalable data collection, cross-domain data fusion, and automated dataset generation.” How is scalability demonstrated or evaluated in the paper?

Response. Scalability was not evaluated in this study, so we removed it from the Abstract and from the claimed contributions. Kafka remains in the architecture as an implemented telemetry transport component, without a scalability conclusion.

Change in the manuscript. The Abstract, framework scope, telemetry section, and Conclusions no longer claim demonstrated scalability.

8. Section IV-A provides an extensive description of the data-acquisition system, but several issues should be addressed. First, the respective roles of InfluxDB and PostgreSQL are repeated in Section III and Sections IV-A.2 and IV-A.4; this implementation logic should not be described excessively. Second, this part of the system appears to be inherited from the authors’ previous work, but the manuscript does not clearly distinguish the innovations of the present study from inherited components. If the entire acquisition system comes from previous work, the space devoted to it is disproportionate to that devoted to the cross-domain semantic modules introduced in this paper. Third, the manuscript includes excessive descriptions of the generic capabilities of third-party components. For example, a general introduction to Kafka is not an author contribution and should be shortened or removed. Finally, the authors should focus on functions that have actually been implemented rather than mixing implemented capabilities with future or potential ones. For example, have the following functions already been implemented: “In addition, lightweight local AI/ML modules can be deployed on distributed databases to support local inference, such as short-term

performance forecasting and anomaly detection” in Section IV-A.2, and “It also supports on-demand aggregation, where only task-relevant telemetry streams are forwarded for cross-domain analysis, optimisation, or orchestration” in Section IV-A.3? The authors should review the manuscript and clearly distinguish implemented functions from prospective capabilities throughout.

Response. We now state the inheritance and additions once at the beginning of Section IV-A. The previous optical platform implemented acquisition, InfluxDB/PostgreSQL storage, Kafka transfer, OpenFaaS AI services, and SDN-controller interfaces. The cited field trial reported database-to-AI transfer below 0.05 s, multi-link ANN QoT inference below 0.3 s, and controller-driven Voyager reconfiguration. The present study adds wireless telemetry, semantic processing of exported records, the unified optical–wireless training interface, and the DT data-generation workflow. Generic Kafka/database explanations and unsupported local-inference statements were removed.

Change in the manuscript. Section IV-A now distinguishes the previously implemented platform from the additions in this study, gives concise evidence for the inherited AI/controller functions, and describes only the data path needed to understand the released datasets.

9. Section IV-B requires more detail. In Section IV-B.1, the authors state that eight semantic features are used. From which raw features are they calculated, how are they calculated, and why were these particular features selected? Does this step correspond to the “task-oriented semantic communication” discussed in Section II-B? The authors are advised to provide a more detailed description of this part of the work.

Response. We specify the 12 input features derived from 30-s wireless records, training-only activity thresholds and scalars, the four targets (load, access diversity, stability, and congestion risk), and the exact equations for diversity, stability, and risk. V1 transmits eight deterministic indicators covering traffic intensity, temporal stability, multi-access availability, signal quality, inactivity, and composite risk. V2/V3 encode the same task context through learned bottlenecks.

Change in the manuscript. Section IV-B.1 now gives the formulas, architecture sizes, optimisation settings, confidence definition, and rationale.

Evidence and boundary. The released material records exact seeds, preprocessing parameters, source fields, and per-seed outputs. The confidence value is a heuristic, not a calibrated probability.

10. Why is V3 described as “privacy-aware”? Does it introduce a substantive improvement over V2, or is the term used simply because three dimensions expose less information than eight dimensions and the added noise makes reconstruction more difficult?

Response. We use privacy-aware to denote a design and evaluation objective, not a formal guarantee. V3 varies bottleneck dimension and training-time noise; attribute inference is evaluated with logistic regression, random forest, MLP, and histogram gradient boosting against raw, V1/V2, PCA, random-projection, majority, and random baselines. V3 moderately reduces observed CPE leakage relative to raw telemetry while maintaining V1-level binary utility, but PCA leaks less and also loses utility. V3 is not universally best.

Change in the manuscript. The semantic introduction, results, Fig. 5, Table 1, and Conclusions now use the bounded empirical interpretation.

11. The sources of the optical-link signals are clearly described, whereas the wireless-signal source is identified only as “a wireless access network.” Are the wireless signals converted from signals on one of the optical links, as suggested by Figs. 1 and 6?

Response. No. The wireless telemetry comes from the 5G-CLARITY/REASON multi-access testbed and is not converted from either optical link. The optical and wireless datasets enter the cross-domain workflow through the unified data interface during subsequent processing and training.

Change in the manuscript. The practical wireless subsection now identifies the wireless testbed and its 5G, Wi-Fi, and LiFi configuration explicitly.

12. The overall multi-agent architecture appears to follow an orchestrator-executor pattern, yet the subagents (e.g., the planner and report agents) seem to operate in a fixed sequence. Is the role and placement of the orchestrator consistent with a conventional multi-agent architecture? Typically, an orchestrator receives the user request and then plans and dispatches tasks to subagents. Why is it positioned in the lower-right corner of the figure? The arrows beneath the first three agents appear to indicate direct communication with those agents. Does the orchestrator operate only for the final two agents, or do those arrows instead represent communication with the Agentic AI layer as a whole? The current diagram does not make this architecture clear.

Response. The orchestrator receives the request and owns the shared task state. It invokes planning, scenario expansion, execution, analysis, reflection, and reporting in sequence. The arrows beneath the roles denote access to that shared orchestration layer; they do not mean that the orchestrator serves only the last two roles. The lower-right position is a control-plane layout choice rather than execution order.

Change in the manuscript. The revised Fig. 6 distinguishes user input, role sequence, shared state, DT tool, and retained output. The article retains only the operational sequence; this layout-level explanation is provided here. The figure is now cropped to its content and displayed near full text width.

13. Figs. 1 and 6 are structurally very similar, which limits the additional information conveyed by Fig. 6. Moreover, why are the AI engines and network controllers absent from Fig. 6? Was the architecture revised, or were the functions of these two components absorbed by other modules in the figure? Fig. 1 and Section III also state that “AI engines provide ML and optimisation capabilities for data-driven inference” and that “Finally, the network controllers translate the optimised decisions or recommendations generated by the Agentic AI layer into executable network configurations, including path provisioning, cross-domain orchestration, and policy-based closed-loop control.” However, these functions are not clearly demonstrated in the discussions or results presented in Sections IV and V. Please provide supporting results for these claimed capabilities.

Response. The two figures now serve different purposes. Fig. 1 shows the complete system architecture and the connections among practical telemetry, repositories, semantic processing, DT generation, AI services, and controllers. Fig. 6 shows the internal sequence of the new orchestrated data-generation workflow, so it includes the shared task state, role-specific calls, simulator interface, and retained artefacts rather than repeating the AI/controller path. The AI engine and SDN controller were implemented and experimentally demonstrated in our earlier optical telemetry platform; Section IV-A now cites that work and reports its measured transfer, inference, and reconfiguration times. The present experiments extend the platform’s data path and evaluate semantic fusion and DT-based generation.

Change in the manuscript. The Fig. 1 and Fig. 6 captions and Sections III–V now state the distinct architectural and workflow roles. Section IV-A provides the published implementation evidence for the AI-engine and controller blocks.

14. Are the five “agents” in Fig. 6 merely nodes that invoke an LLM, or are they autonomous agents? If they are agents, their internal structures should be incorporated into Fig. 6, including their tools, memory-management mechanisms, and intermediate scheduling components. Although a multi-agent architecture can improve flexibility and completeness, it also introduces inter-agent communication overhead, increases the number of LLM calls, and makes the overall framework less controllable. The authors should therefore explain why the Planner Agent, Scenario Expansion Agent, and Report Agent are implemented as agents when their assigned tasks do not appear particularly complex. If these components are not agents, the relevant descriptions should be revised and the characterization of the framework as multi-agent should be reconsidered.

Response. We retain the multi-agent architecture as the experimental design and give an operational definition: the five agents are schema-defined functional roles coordinated through orchestrator-owned shared state. Planner, Scenario Expansion, and Reflection invoke the LLM; Results Analysis and Report are deterministic. Simulator wrappers, topology/equipment files, KPI definitions, and stored results are their tools and knowledge. Role separation provides explicit input/output contracts, traceable intermediate artefacts, and deterministic replacement points. The roles exchange serialised state through the orchestrator rather than peer-to-peer messages. The instrumented run quantifies the observable LLM overhead as six calls, 108,308 tokens, and 26.22 s summed API latency; it does not isolate an inter-role communication ablation or establish an efficiency advantage.

Change in the manuscript. Fig. 6 and Section V-A now show the operational role definitions, shared state, tools, retained artefacts, and orchestration sequence. Table 3 reports the measured call, token, latency, cost, retry, and failure totals.

15. In Fig. 6, the font color used for the Reflection Agent is too similar to the background color, making the text difficult to read. Please avoid this color combination.

Response. Corrected as requested. The current high-resolution asset uses darker Reflection text and improved contrast.

16. Fig. 7 is not sufficiently clear and should be provided at a higher resolution.

Response. Fig. 7 was regenerated from the corrected unique configuration table and is included as a vector PDF. Its caption distinguishes execution records, unique configurations, valid configurations, and all-off cases.

Reviewer 3

1. The semantic-fusion experiments are not reproducible. The manuscript does not identify the optical and wireless data used or explain how the fused dataset was constructed. Please provide sample counts, label definitions, semantic confidence scores, model architectures, training settings, and the evaluation protocol. Random record-level splitting may leak correlated samples; use temporal or session-grouped splits.

Response. We replaced the random split with chronological 60/20/20 train/validation/test segments and ten-record purge gaps, yielding 1,698/560/570 evaluated samples. All thresholds, scalars, encoders, classifiers, and attackers are fitted after splitting using training data only. The manuscript now defines the 12 wireless inputs, semantic features, confidence heuristic, optical and wireless risk rules, one-step targets, model architectures, optimiser settings, ten seeds, and evaluation metrics.

Change in the manuscript. Section IV-B and the public materialised 2,848-row/52-field benchmark now record source indices, split membership, thresholds, features, and labels.

Evidence and boundary. The optical and wireless records were collected in different time periods because of experimental-platform scheduling. The released benchmark table records source-row indices, unified-interface outputs, split membership, and rule-derived labels.

2. Table 1 does not establish improved privacy. V3 has the highest RAT attack accuracy (0.988), while its lower Macro-F1 cannot be interpreted without class distributions and baselines. Please add majority and random baselines, attacker performance on raw telemetry, multiple attacker models, repeated runs, and confidence intervals. Until then, use “privacy-aware” rather than “privacy-preserving.”

Response. We added all requested comparisons. CPE Macro-F1 is the primary privacy metric because the RAT test set is highly imbalanced. Four attacker families are evaluated over ten seeds. Raw, PCA, random projection, majority, and stratified-random baselines are included, with 95% confidence intervals. V3 achieves strongest-observed CPE attacker Macro-F1 0.2382 ± 0.0113 , compared with 0.2757 for raw features; PCA3 reaches 0.2174 ± 0.0059 but with lower utility.

Change in the manuscript. The manuscript consistently uses “privacy-aware” and states that there is no formal or universal privacy guarantee.

3. The LLM framework is demonstrated but not evaluated. The GNPpy parameter space could be enumerated by a deterministic script, yet no baseline is provided. Report the LLM model, prompts, settings, trials, success rate, reflection iterations, human interventions, failures, runtime, and token cost. Alternatively, limit the claims to a feasibility demonstration.

Response. We selected the suggested feasibility scope. The full instrumented-run details and deterministic baseline are reported under R1.2. The article retains the model/settings, six-call summary, cost/latency, baseline, and manual-recovery boundary. Complete prompts, responses, failure logs, and intervention chronology are released but moved out of the main narrative.

Evidence and boundary. The final Reflection response requested another refinement at the configured two-iteration limit and was not recomputed after the corrected replay; convergence and autonomous repair are therefore not claimed. No repeated LLM trial or role ablation is claimed.

4. The generated data lack independent correctness validation. Runtime checks may catch execution failures but not plausible results from incorrect configurations. Add an audit using independently reproduced samples, a hand-written generator, established GNPpy examples, or field measurements. Identify the repository commit and report simulator versions and material configuration assumptions.

Response. For GNPpy, 96 cases stratified by path, joint modulation/slot-width setting, power, and channel load were regenerated independently. Official GNPpy v2.12 (commit `7ce665010970f57e46672db1ec0864ae448077e6`) completed 96/96. Its mean and maximum absolute GSNR differences from the archive were 0.00015 and 0.00050 dB. A schema-corrected GNPpy v2.14.2 run completed 96/96, with corresponding differences of 0.628 and 1.435 dB, demonstrating version sensitivity. Both arms retain EDFA/ROADM warnings; “successful” therefore means numerically reproducible under declared assumptions, not physically calibrated.

The corrected refined-pass audit contains 6,144 unique cases: 6,120 active scenarios with matching power in scenario metadata, request JSON, CLI arguments, and returned metrics, plus 24 expected all-off cases. Its 6,120 result files contain 48,960 EDFA minimum-gain warnings and 12,240 unmet

ROADM target-power warnings. These exact counts and configuration repairs are retained in the repository rather than the article.

For wireless data, independent LiFi equations pass 9/9 tests; compiled ns-3 probes agree with an independent Python reference in 15/15 values; one integrated run maps all nine application flows; same-seed outputs are byte-identical; and the 48-configuration sweep has 432/432 application mappings. These establish implementation consistency under the stated model, not calibration to the physical testbed.

Change in the manuscript. Section V-C and Tables 5–6 report the essential versioned replay and wireless validation results; forensic counts and configuration changes are documented here and in the public artefacts.

5. Separate implemented, evaluated, demonstrated, and proposed components in the abstract.

Response. The revised Abstract distinguishes the implemented cross-domain architecture and unified interface, the separately timed practical acquisitions, the evaluated semantic-fusion benchmark, the LLM/DT workflow, the deterministic baseline, and the versioned numerical replay. The inherited AI-engine and controller implementation is described and cited in Sections III and IV-A rather than presented as a new experimental result.

6. Sections 3 and 4.A overlap; shorten Section 3 and use the space for methodology.

Response. Section III now gives the two data workflows, the orchestrator relationship, and the boundary between inherited platform functions and the contributions of this study. Section IV-A contains the component-level implementation details and the evidence for the inherited AI-engine/controller services. The repeated database, telemetry, and capability descriptions were removed.

7. Explain the relevance of the AC/DC power-supply detail or remove it.

Response. Removed as requested because it is not needed for the dataset or DT claims.

8. Support or soften the claim that -5 dBm provides optimal performance.

Response. We now call -5 dBm the operating point used for the practical campaign. The synthetic GNPpy results are described as route-dependent sensitivity over three tested powers; no global optimum is claimed.

9. Standardise British or American English throughout the manuscript. The text currently mixes forms such as “centralised/centralized,” “optimisation/optimization,” “behaviour/behavior,” and “fibre/fiber.” Select one spelling convention and apply it consistently...

Response. We selected British English and standardised active prose (for example, organised, optimisation, behaviour, fibre, centralised). Bibliographic titles and software identifiers retain their source spelling.

10. Use “ns-3” consistently and standardise heading capitalisation.

Response. Corrected throughout. The final synthetic backend is identified as ns-3.44 with CTTC 5G-LENA NR, native Wi-Fi 802.11ax, and an equation-driven LOS LiFi/OWC model. Internal development labels were removed from reader-facing prose except where they form stable repository paths.

11. Replace or remove the placeholder DOI on page 1.

Response. Removed. The journal class inserts `http://dx.doi.org/10.1364/ao.XX.XXXXXX` by default, even when the template DOI line in the manuscript is commented. It also inserts `Compiled \today` through its default date field. We now explicitly set both `\doi{}` and `\dates{}`, rebuild the article, and verify the first-page PDF text. The publisher can insert the assigned DOI and publication dates after acceptance.

12. Correct “includign BER,” “based over the NDFP topology,” and “formats,i.e.”.

Response. Corrected as requested. We also performed a final mechanical check of active prose, citations, references, terminology, and heading capitalisation.

Availability of revision evidence

The revised manuscript source, compiled PDF, figures, response letter, Croissant metadata, semantic evaluation outputs, deterministic baseline, GNPY replay, LLM trace, and synthetic wireless validation artefacts are collected in the cited revision repositories. The datasets and associated metadata/documentation are licensed under CC BY-NC 4.0, and the article uses “openly available” consistently with those terms. Software and cited external datasets retain their stated licences; simulator-calibration and workflow-trace boundaries are documented with the corresponding evidence.